

A Four-Prong Obstruction to (3+1)-Dimensional Emergence from $\text{Cl}(6)$

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Abstract

We prove that the $\text{Cl}(6)$ bivector algebra $\mathfrak{so}(6)$, viewed as the internal symmetry algebra of a $\text{Cl}(6)$ -based emergence programme, admits no clean assignment of spatial rotation generators compatible with the simultaneous preservation of $\mathfrak{su}(3)$ color. The obstruction is dimension-theoretic and basis-independent: any designation of two internal directions as spatial leaves a four-direction complement whose bivector algebra is $\mathfrak{so}(4)$, of dimension six, which is strictly less than the eight required to house $\mathfrak{su}(3)$. Combined with three further algebraic obstructions—the one-dimensional centralizer of $\mathfrak{su}(3)$ in $\mathfrak{so}(6)$, the one-dimensional nature of the chain axis that supplies the 1+1 sector, and the parallel double-booking dilemma on the quaternion factor $\mathbb{H}$ —the result constitutes a four-prong structural no-go for (3+1) emergence under the full Stance-1 hypothesis (no imported spacetime or gauge structure). All four prongs are computationally verified at machine precision. The no-go sharpens and completes earlier partial results. Under Stance 1 the only surviving escape is whether the emergent infrared theory voids the hypotheses of the Coleman–Mandula theorem by being genuinely gapless with no asymptotic particle states; this remains open. Under Stance 2, the centralizer theorem provides an algebraic justification for every import rather than a default assumption.

1 Introduction

The TMFT emergence programme asks whether a (3+1)-dimensional spacetime containing the Standard Model gauge group can be derived, without import, from the Clifford algebra $\text{Cl}(6)$ tensored with $\text{Cl}(1, 3)$. The C-series

lattice construction has successfully produced a clean 1+1 Lorentzian sector: a massless chiral fermion along an external chain axis n , with the Tomita–Takesaki modular structure supplying the 1+1 Lorentzian signature (Firestone, 2026a). The 1+1 result is established. The question addressed here is whether the same construction, or any algebraic extension of it within the same $\text{Cl}(6)$ structure, can produce two additional spatial directions and recover (3+1).

The $\text{Cl}(6)$ bivector algebra is $\mathfrak{so}(6)$, of dimension 15. Its maximal $\mathfrak{su}(3)$ -color subalgebra occupies eight of those fifteen dimensions. A prior result (Firestone, 2026f) established that the centralizer of $\mathfrak{su}(3)_{\text{color}}$ in $\mathfrak{so}(6)$ is exactly one-dimensional, spanned by $H = B_{12} + B_{34} + B_{56}$, which is proportional to the baryon-minus-lepton charge Q_{B-L} . That single result has four distinct structural consequences, each closing a different candidate route to (3+1). Together they form a complete no-go theorem.

The paper is organized as follows. Section 2 states and verifies the centralizer theorem. Section 3 derives the four consequences. Section 4 states the partition-double-booking dilemma and its dimension-theoretic proof. Section 5 closes the quaternion spatial route. Section 6 assembles the four-prong no-go. Section 7 identifies the two surviving paths: the beta question (whether the emergent infrared voids Coleman–Mandula’s hypotheses by being genuinely gapless with no asymptotic particle states) and Stance 2 (imported spacetime and gauge structure, the fallback if beta fails, with each import retrospectively justified by the centralizer theorem rather than assumed by default). Section 8 discusses the structure of the no-go and why all four prongs are necessary.

Conventions. $\text{Cl}(6)$ is generated by $e_1, \dots, e_6$ with $\{e_i, e_j\} = 2\delta_{ij}$, all e_i Hermitian. Bivectors $B_{ij} = \frac{1}{2}[e_i, e_j]$ span $\mathfrak{so}(6)$. Throughout, the candidate symmetry algebra is the Lie algebra of bivectors $\mathfrak{so}(6)$, which generates the spin group $\text{Spin}(6)$ by exponentiation. This is the standard choice in geometric-algebra physics: gauge and spacetime symmetries act through the bivector algebra. Higher-grade elements of $\text{Cl}(6)$ (4-vectors, 6-vectors, the volume element), when exponentiated, generate extended groups (conformal or higher-graded transformations) outside the $\text{Spin}(p, q)$ gauge and spacetime framework of the present construction, and are excluded on those grounds. The $\mathfrak{su}(3)$ subalgebra is derived as the stabilizer of the $H = -3/2$ eigenstate of the number operator. The stabilizer construction of $\mathfrak{su}(3)$ inside $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ goes back to Günaydin and Gürsey (1973) and was elaborated by Dixon (1994); the formulation used here follows Furey (2016). The

quaternion factor $\mathbb{H}$ acts on a four-dimensional real space ($\mathbb{H}$ as real vector space) via left and right multiplication; $\mathfrak{su}(2)_L = \text{span}\{L_i/2, L_j/2, L_k/2\}$ and $\mathfrak{su}(2)_R = \text{span}\{-R_i/2, -R_j/2, -R_k/2\}$.

2 The Centralizer Theorem

Theorem 1 (Centralizer). *The centralizer of $\mathfrak{su}(3)_{\text{color}}$ inside the $\text{Cl}(6)$ bivector Lie algebra $\mathfrak{so}(6)$ is exactly one-dimensional:*

$$C_{\mathfrak{so}(6)}(\mathfrak{su}(3)) = \text{span}\{H\}, \quad H = B_{12} + B_{34} + B_{56} = \frac{3}{2} Q_{B-L}. \quad (1)$$

No other bivector commutes with all eight color generators.

Proof. The result follows from Schur's lemma, basis-independently. The $\text{Cl}(6)$ bivector algebra is $\mathfrak{so}(6) \cong \mathfrak{su}(4)$, acting on $\mathbb{C}^4$. The $\mathfrak{su}(3)_{\text{color}}$ subalgebra is the Furey embedding $\mathfrak{su}(3) \subset \mathfrak{su}(4)$, under which $\mathbb{C}^4$ decomposes as the reducible $\mathbf{3} \oplus \mathbf{1}$: the fundamental $\mathbf{3}$ of $\mathfrak{su}(3)$ plus a singlet. By Schur's lemma, an element of $\text{End}(\mathbb{C}^4)$ commuting with all of $\mathfrak{su}(3)$ acts as a scalar on each irreducible block, so the commutant of $\mathfrak{su}(3)$ in $\text{End}(\mathbb{C}^4)$ is $\mathbb{C} \oplus \mathbb{C}$, two complex-dimensional, spanned by the two block projectors. Restricting to $\mathfrak{su}(4)$ (traceless Hermitian, in the physics convention where $\mathfrak{su}(4)$ generators are Hermitian) reduces the four real dimensions of the $\mathbb{C} \oplus \mathbb{C}$ commutant to one: Hermiticity forces α and β real (removing two), and tracelessness imposes $3\alpha + \beta = 0$ (removing one), leaving one real dimension. The surviving generator is proportional to $\text{diag}(1, 1, 1, -3)$, which is exactly the hypercharge / Q_{B-L} assignment. The dimension of the centralizer is independent of basis and of representation; the identification with the specific bivector combination $H = B_{12} + B_{34} + B_{56}$ fixes the bivector basis to the canonical mode-pair decomposition $(e_1e_2), (e_3e_4), (e_5e_6)$. $\square$

The script `cl6_su3_centralizer_check.py` corroborates this in the explicit 8-dimensional spinor representation: the centralizer kernel is one-dimensional with alignment $|\langle v, H \rangle| / \|v\| \|H\| = 1.0000000000$ at machine precision, and the $\mathfrak{su}(3)$ octet closes at residual 6.2×10^{-16} .

Lemma 2 (Decomposition). *Every bivector decomposes across three orthogonal $\mathfrak{su}(3)$ sectors—adjoint-8, centralizer-1, and coset- $(\mathbf{3} + \bar{\mathbf{3}})$ —as follows:*

- Each within-pair Cartan element B_{12}, B_{34}, B_{56} has exactly $2/3$ adjoint content and $1/3$ centralizer content, zero coset.
- Each cross-plane bivector has exactly $1/2$ adjoint content and $1/2$ coset content, zero centralizer.

- *No bivector lies purely in any single sector.*

The fractions in Lemma 2 are not numerical accidents. The three-dimensional Cartan space $\text{span}\{B_{12}, B_{34}, B_{56}\}$ decomposes as $\text{span}\{h_1, h_2\}$ (two-dimensional $\mathfrak{su}(3)$ Cartan) $\oplus \text{span}\{H\}$ (one-dimensional centralizer); each Cartan bivector therefore carries 2/3 adjoint and 1/3 centralizer by a simple projection count. The cross-plane result follows from the absence of H -content in off-diagonal bivectors, which is forced by the coset structure.

Corollary 3. *Within $\mathfrak{so}(6)$, exactly one independent $U(1)$ commutes with $\mathfrak{su}(3)_{\text{color}}$ —the one generated by $H = \frac{3}{2} Q_{B-L}$. Any structure requiring more than one commuting generator, or requiring a commuting generator free of $B-L$ charge, cannot be housed in the $\text{Cl}(6)$ bivector algebra.*

3 Four Consequences of One Theorem

The single fact that $\dim C_{\mathfrak{so}(6)}(\mathfrak{su}(3)) = 1$ closes four independent candidate assignments. We take each in turn.

3.1 Hypercharge—the seat that fits

$U(1)_Y$ hypercharge requires one generator commuting with $\mathfrak{su}(3)_{\text{color}}$. Theorem 1 provides exactly one: H . Identifying $Y \propto H$ (Paper 2: $Q_{B-L} = (2/3)H$) places hypercharge in the unique color-neutral bivector. The centralizer being one-dimensional explains why there is precisely one independent commuting $U(1)$ —not zero, not two. Hypercharge takes that seat completely.

3.2 Weak isospin $SU(2)_L$ —locked out of $\text{Cl}(6)$ bivectors

$SU(2)_L$ requires three generators commuting with $\mathfrak{su}(3)_{\text{color}}$. The centralizer has dimension one. Three independent generators cannot fit.

This provides the algebraic resolution of the B-2 finding from the Paper 3 adversarial referee review (June 2026): the corrected $T_3 = -B_{34}$ fails to commute with the color $\mathfrak{su}(3)$ because B_{34} is 2/3 color-adjoint (Lemma 2). The B-2 finding is the centralizer theorem asserting itself on the electroweak sector. No choice of T_1, T_2, T_3 from within the $\text{Cl}(6)$ bivector algebra can commute with all of $\mathfrak{su}(3)_{\text{color}}$. $SU(2)_L$ must be accommodated on the separate $\mathbb{H}$ factor of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ —the standard Furey construction (Furey, 2016).

3.3 Spatial $\mathfrak{so}(3)$ —locked out (two mechanisms)

Spatial rotation $\mathfrak{so}(3)$ requires three generators. The centralizer route needs $\dim \geq 3$ in $C_{\mathfrak{so}(6)}(\mathfrak{su}(3))$; Theorem 1 gives $\dim = 1$. This closes the double-booking route. The partition route is closed separately and is the subject of Section 4.

3.4 Color-neutral photon helicity—locked out (B – L reduction)

A color-neutral helicity generator would have to be proportional to H (Corollary 3). But $H = \frac{3}{2} Q_{B-L}$ carries baryon-minus-lepton charge. Photons carry $B-L = 0$. The contradiction is immediate: transverse polarization modes cannot be color-neutral within the $\text{Cl}(6)$ bivector algebra. This is the algebraic half of the two-walls identity established in (Firestone, 2026e).

3.5 Summary

Sector	Generators needed	Fits in centralizer ($\dim 1$)?	Verdict
$\text{U}(1)_Y$ hypercharge	1	Yes— $Y \propto H$	FITS
$\text{SU}(2)_L$ weak isospin	3	No— $3 > 1$	LOCKED OUT
Spatial $\mathfrak{so}(3)$	3	No— $3 > 1$ (double-booking)	LOCKED OUT
Color-neutral helicity	1, B – L -free	No— H carries B – L	LOCKED OUT

Table 1: Centralizer assignments. One seat. Four candidates. Only hypercharge fits.

4 The Partition-Double-Booking Dilemma

The C-series construction produces a 1+1 Lorentzian sector along an external chain axis n (Firestone, 2026a). The chain axis is external—it does not come from the $\text{Cl}(6)$ internal directions. For (3+1), two additional spatial directions must be produced. The only candidate source within the current construction is the six-dimensional space spanned by $e_1, \dots, e_6$. We now show that no assignment of two of these directions as spatial is consistent with the survival of $\mathfrak{su}(3)_{\text{color}}$.

Theorem 4 (Partition No-Go). *Let V_2 be any two-dimensional subspace of $\mathbb{R}^6$. Its orthogonal complement V_4 has bivector algebra $\mathfrak{so}(V_4)$, with*

$$\dim \mathfrak{so}(V_4) = \binom{4}{2} = 6 < 8 = \dim \mathfrak{su}(3). \quad (2)$$

Therefore no eight-dimensional Lie algebra, $\mathfrak{su}(3)$ in particular, is a subalgebra of $\mathfrak{so}(V_4)$. Under any partition of the six $\text{Cl}(6)$ directions into two spatial and four internal, $\mathfrak{su}(3)_{\text{color}}$ cannot survive as a subalgebra of the residual internal bivector algebra.

Proof. The bivector count is purely combinatorial. The dimension of $\mathfrak{so}(V_4)$ is the number of 2-element subsets of a 4-element basis: $\binom{4}{2} = 6$. Since $\mathfrak{su}(3)$ has dimension 8 and $8 > 6$, no Lie algebra isomorphic to $\mathfrak{su}(3)$ can be a subalgebra of the six-dimensional $\mathfrak{so}(V_4)$. This is independent of the choice of V_2 . $\square$

Numerical concretization. Theorem 4 is the basis-independent statement; the script `cl6_spatial_direction_carving.py` concretizes it in a fixed basis by recording, for each partition, how many of the eight original $\mathfrak{su}(3)$ generators remain intact (free of spatial-sector components) in the residual. This is a stronger per-case fact than the theorem requires—the theorem only needs that no $\mathfrak{su}(3)$ subalgebra exists in $\mathfrak{so}(V_4)$ —but the per-case leakage count corroborates it concretely. The script tests all inequivalent assignment types (two cases up to S_3 permutation symmetry of the mode pairs, plus all eight cross-pair variants exhaustively):

- **Case A (within-pair):** spatial directions e_5, e_6 . Spatial rotation generator B_{56} . Six purely internal bivectors span $\mathfrak{so}(4)$ at closure residual 3.14×10^{-16} ; $\mathfrak{su}(3)$ survival = 1/8 generators intact. H disrupted at $\|H - \text{proj}\|/\|H\| = 0.577$. $\mathfrak{su}(3)$ **broken**.
- **Case B (cross-pair):** spatial directions e_3, e_5 . Spatial rotation generator B_{35} . Six purely internal bivectors span $\mathfrak{so}(4)$ at closure residual 3.14×10^{-16} ; $\mathfrak{su}(3)$ survival = 0/8 generators intact. H disrupted at $\|H - \text{proj}\|/\|H\| = 0.816$. $\mathfrak{su}(3)$ **broken**.
- **All eight cross-pair variants:** every assignment yields six internal bivectors, dimension six, 0/8 $\mathfrak{su}(3)$ generators surviving. The result is uniform across all choices.

The dilemma. The double-booking obstruction (Sections 2–3, consequence (iii)) and the partition no-go (Theorem 4) together form a dilemma with no middle ground:

- **If you partition** (designate two directions spatial, keep four internal): $\mathfrak{su}(3)$ is destroyed in the residual sector by Theorem 4.
- **If you do not partition** (all six directions serve dual spatial and color roles—double-booking): spatial rotations $\mathfrak{so}(3)$ fail to commute with color $\mathfrak{su}(3)$, since $C_{\mathfrak{so}(6)}(\mathfrak{su}(3)) = \text{span}\{H\}$ is one-dimensional and cannot house the three generators $\mathfrak{so}(3)$ requires (Theorem 1, consequence (iii)).

The six internal directions are the only reservoir. Either partition them or do not. Each horn is fatal by a different mechanism.

5 The Quaternion Spatial Route—Closed

The dilemma of Section 4 operates entirely within the $\text{Cl}(6)$ bivector algebra. A natural candidate escape is to seek spatial rotation generators from the quaternion factor $\mathbb{H}$ of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, particularly since $\text{SU}(2)_L$ is already assigned there (Section 3). We now show that $\mathbb{H}$ provides no clean spatial route either.

Theorem 5 ($\mathbb{H}$ Dilemma). *The quaternion algebra $\mathbb{H}$, viewed as a four-dimensional real vector space with left and right multiplication algebras*

$$\mathfrak{su}(2)_L = \text{span}\{L_i/2, L_j/2, L_k/2\}, \quad (3)$$

$$\mathfrak{su}(2)_R = \text{span}\{-R_i/2, -R_j/2, -R_k/2\}, \quad (4)$$

satisfies:

$$C_{\text{span}\{L_a, R_b, I\}}(\mathfrak{su}(2)_L) = \mathfrak{su}(2)_R \oplus \mathbb{R} \quad (\dim 4), \quad (5)$$

$$C_{\text{span}\{L_a, R_b, I\}}(\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R) = \text{span}\{I\} \quad (\dim 1). \quad (6)$$

The scalar identity I is not a rotation generator. Therefore no non-trivial generator in $\mathbb{H}$ commutes with both internal $\text{SU}(2)$ factors.

Proof. Direct computation (script `cl6_H_spatial_route_check.py`):

- $\mathfrak{su}(2)_L$ closes: $[L_a, L_b] = 2\varepsilon_{abc} L_c$, residual $0.00 \times 10^{+00}$. [CHECK A]

- $\mathfrak{su}(2)_R$ closes: $[R_a, R_b] = -2\varepsilon_{abc} R_c$, residual $0.00 \times 10^{+00}$; $S_a = -R_a/2$ satisfies $[S_a, S_b] = \varepsilon_{abc} S_c$ (genuine $\mathfrak{su}(2)$). [CHECK B]
- Cross-commutativity: $[L_a, R_b] = 0$ for all nine pairs, $0.00 \times 10^{+00}$. [CHECK C]
- Centralizer of $\mathfrak{su}(2)_L$: basis $\{I, R_i, R_j, R_k\}$, dim 4, zero L_a content. [CHECK D]
- $\mathfrak{su}(2)_R$ is nonabelian: $\max \|[R_a, R_b]\| = 4.0$ (not a $U(1)$). [CHECK E]
- Centralizer of $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$: basis $\{I\}$, dim 1, identity fraction 1.000000. [CHECK F]

□

The $\mathbb{H}$ dilemma. The quaternion factor $\mathbb{H}$ is the internal factor of the $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ decomposition by construction; every element of its rotation algebra $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is therefore internal, including $\mathfrak{su}(2)_L$ (weak isospin), $\mathfrak{su}(2)_R$, the diagonal $\mathfrak{su}(2)_{\text{diag}} = \text{span}\{(L_a - R_a)/2\}$, and any other three-dimensional subalgebra. A spatial rotation $\mathfrak{so}(3)$ would have to commute with the entire internal $\mathbb{H}$ -structure to avoid double-booking. The load-bearing fact is the centralizer: $C(\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R) = \text{span}\{I\}$ [Check F]. Any element of the $\mathbb{H}$ rotation algebra that commutes with the full internal structure must be proportional to the scalar identity I , and I is not a rotation generator ($\exp(\theta I)$ is a dilation of $\mathbb{R}^4$, not a rotation; I is symmetric, not antisymmetric, hence not in $\mathfrak{so}(4)$). No three-dimensional spatial rotation algebra can be sourced from $\mathbb{H}$.

This subsumes the individual subalgebra cases. The candidate $\mathfrak{su}(2)_{\text{diag}}$, for instance, does not commute with $\mathfrak{su}(2)_L$: $[L_a - R_a, L_b] = [L_a, L_b] = 2\varepsilon_{abc} L_c$ is nonzero for $a \neq b$ (using cross-commutativity $[R_a, L_b] = 0$, Check C). The same computation rules out any other would-be spatial subalgebra by the centralizer result directly, with no need to enumerate cases.

The obstruction is algebraic and complete at this level. The Coleman–Mandula confrontation enters only conditionally, as in prong (ii): if the emergence programme reaches a 4D theory meeting Coleman–Mandula’s hypotheses (Section 7), mixing Poincaré spatial rotations with any internal symmetry on $\mathbb{H}$ is forbidden there as well, independently of the tensor factor on which the mixing occurs. The algebraic obstruction does not need Coleman–Mandula; Coleman–Mandula is the backup that closes the same door at the level of the target theory (Coleman and Mandula, 1967).

6 The Four-Prong No-Go

We assemble the results of the preceding sections.

Theorem 6 (Four-Prong No-Go). *Under Stance 1 (no imported spacetime or gauge structure), the $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ algebraic construction admits no clean (3+1)-dimensional extension of the established 1+1 sector. Four independent obstructions hold simultaneously:*

(i) **Partition kills $\mathfrak{su}(3)_{\text{color}}$.** Any designation of two of the six $\text{Cl}(6)$ internal directions as spatial leaves a four-direction complement with bivector algebra $\mathfrak{so}(4)$, $\dim 6$, which is too small to house $\mathfrak{su}(3)$ ($\dim 8$). [Theorem 4, ESTABLISHED]

(ii) **Double-booking triggers Coleman–Mandula.** Not partitioning forces spatial $\mathfrak{so}(3)$ to share generators with color $\mathfrak{su}(3)$, since the centralizer $C_{\mathfrak{so}(6)}(\mathfrak{su}(3))$ is one-dimensional (Theorem 1) and cannot house the three independent generators $\mathfrak{so}(3)$ requires. Coleman–Mandula’s conclusion, applied to the would-be 4D emergent theory under the standard hypotheses (Poincaré-invariant interacting QFT with a mass gap and a nontrivial S -matrix, stated in full in Section 7), is that the symmetry algebra is a direct sum of the Poincaré algebra and an internal algebra commuting with it: $[\text{Lorentz}, \text{internal}] = 0$. A spatial rotation generator that shares with $\mathfrak{su}(3)_{\text{color}}$ would then be required to commute with all of Lorentz while also having nonzero brackets with the other $\mathfrak{su}(3)$ generators ($\mathfrak{su}(3)$ is non-abelian); these cannot both hold. The invocation is therefore conditional: if the emergence programme reaches a 4D theory meeting Coleman–Mandula’s hypotheses, that theory cannot carry the forced spatial-color sharing, so the target is unreachable under those hypotheses. The escape is to void the hypotheses (Branch A, Section 7). (Coleman and Mandula, 1967; Haag et al., 1975).

(iii) **The chain gives only 1+1.** The external chain axis n provides the 1+1 Lorentzian sector and is one-dimensional. Three mechanisms block extension: (a) multiple Fermi points on a 1D chain yield parallel translations along n only—different group velocities, same direction; (b) the per-site $M_8(\mathbb{C})$ algebra is type I_8 (finite-dimensional, discrete spectrum) and cannot support the HSMI machinery; (c) the 3-leg ladder reinterpretation of the three $\text{Cl}(6)$ mode-pairs gives a discrete transverse quantum number (3 points, no continuum limit)—the transverse shift is a color permutation (S_3 Weyl element), not a spatial translation. (Firestone, 2026d).

(iv) The $\mathbb{H}$ factor double-books. The quaternion factor $\mathbb{H}$ houses $\mathrm{SU}(2)_L$ (forced by prong (ii); see also Section 3). The $\mathbb{H}$ factor is internal by the $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ decomposition, so its entire rotation algebra $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is internal. The centralizer of that full internal structure is $\mathrm{span}\{I\}$ (Theorem 5), and the scalar I is not a rotation generator; no spatial $\mathfrak{so}(3)$ can be sourced from $\mathbb{H}$.

Cross-tensor combinations. A candidate spatial generator need not lie wholly within one tensor factor; it could combine a $\mathrm{Cl}(6)$ bivector part and an $\mathbb{H}$ rotation part. Such a generator must commute with the full internal symmetry across both factors to avoid double-booking, so its $\mathrm{Cl}(6)$ component must lie in $C_{\mathfrak{so}(6)}(\mathfrak{su}(3)) = \mathrm{span}\{H\}$ (prongs (i), (ii)) and its $\mathbb{H}$ component must lie in the centralizer of $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R = \mathrm{span}\{I\}$ (prong (iv)). The $\mathrm{Cl}(6)$ component then supplies at most one dimension of rotation (the $\mathrm{U}(1)$ generated by H in the internal sector); the $\mathbb{H}$ component contributes no rotation generator at all (I is symmetric, not in $\mathfrak{so}(4)$). Together they span at most a two-dimensional abelian Lie algebra, while spatial $\mathfrak{so}(3)$ requires a three-dimensional nonabelian one. No admissible combination yields $\mathfrak{so}(3)$; cross-tensor combinations reduce to the per-factor obstructions and open no new route.

All four prongs are computationally verified at machine precision across three independent scripts (`cl6_su3_centralizer_check.py`, `cl6_spatial_direction_carving.py`, `cl6_H_spatial_route_check.py`).

Corollary 7 (Verification record). *The following items are verified at machine precision:*

7 What Survives

7.1 The two surviving paths

The four-prong no-go closes all algebraic routes to (3+1) under Stance 1. Two paths survive.

Path A: beta (physical escape). The no-go prongs (ii) and (iv) invoke Coleman–Mandula in the standard form, which requires: (1) a Poincaré-invariant 4D QFT, (2) a nontrivial analytic S -matrix with a mass gap, and (3) a finite number of particle species. The algebraic obstructions established here assume all three hypotheses hold. If the emergent infrared theory violates hypothesis (2)—by being genuinely gapless with no asymptotic

Result	Script	Residual / value
Centralizer of $\mathfrak{su}(3)$ in $\mathfrak{so}(6) = \text{span}\{H\}$, dim 1	<code>centralizer_check</code>	dim = 1 PASS
Alignment of centralizer with H	<code>centralizer_check</code>	$ \langle v, H \rangle = 1.0000000000$
$\mathfrak{su}(3)$ octet closure	<code>centralizer_check</code>	6.20×10^{-16}
Within-pair: 2/3 adjoint +1/3 centralizer	<code>direction_carving</code>	exact fractions
Cross-plane: 1/2 adjoint +1/2 coset	<code>direction_carving</code>	exact fractions
Case A (within-pair spatial): $\mathfrak{su}(3)$ broken	<code>direction_carving</code>	1/8 generators
Case B (cross-pair spatial): $\mathfrak{su}(3)$ broken	<code>direction_carving</code>	0/8 generators
All 8 cross-pair variants: $\mathfrak{su}(3)$ broken	<code>direction_carving</code>	uniform 0/8
$\mathfrak{su}(2)_L$ closure in $\mathbb{H}$	<code>H_spatial_route</code>	$0.00 \times 10^{+00}$
$\mathfrak{su}(2)_R$ closure in $\mathbb{H}$	<code>H_spatial_route</code>	$0.00 \times 10^{+00}$
Cross-commutativity $[L_a, R_b] = 0$	<code>H_spatial_route</code>	$0.00 \times 10^{+00}$
Centralizer($\mathfrak{su}(2)_L$) = $\mathfrak{su}(2)_R \oplus \mathbb{R}$, dim 4	<code>H_spatial_route</code>	dim = 4 PASS
$\mathfrak{su}(2)_R$ nonabelian	<code>H_spatial_route</code>	$\max \ [R, R]\ = 4.0$
Centralizer($\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$) = $\text{span}\{I\}$, dim 1	<code>H_spatial_route</code>	dim = 1 PASS

Table 2: Computational verification record. All residuals at or below machine precision. Script names abbreviated from full `c16_...py` filenames.

particle S -matrix—Coleman–Mandula’s conclusion does not apply before its premises do.

The committed specific escape is Branch A: the emergent IR has no asymptotic-particle S -matrix in the genuine conformal or continuous-spectrum sense. Confinement does not qualify—it relabels asymptotic states as hadrons but retains an S -matrix; the escape requires eliminating asymptotic states entirely. Branch A was committed prior to this paper (Firestone, 2026c); the four-prong no-go sharpens its target by specifying exactly what the emergent IR must avoid producing: a standard Poincaré representation on asymptotic states acting independently of color.

The MZ and AD theorems (Maldacena and Zhiboedov, 2013; Alba and Diab, 2016) are the natural analogues of Coleman–Mandula for CFTs: a unitary CFT in $d \geq 3$ with a conserved higher-spin current of spin > 2 and a unique local stress tensor is necessarily free. Neither theorem engages on the C5-delta 1+1 sector, and the primary reason is dimensional. MZ’s

argument is stated for $d = 3$ and AD extends it to $d > 3$. The $d = 2$ case admits a richer higher-spin structure (W_N algebras, integrable theories) that the proofs of MZ and AD do not engage. The proofs rely on the finite dimensionality of the conformal algebra $\mathfrak{so}(d, 2)$, which fails in $d = 2$ where the conformal symmetry is the infinite-dimensional Virasoro algebra. The 1+1 chiral sector is therefore outside the stated domain of both theorems by dimension alone, and this is the load-bearing escape.

A second structural observation reinforces the dimensional escape but is not required by it. MZ and AD both take a unique local stress tensor as an explicit hypothesis, and both sets of authors explicitly disclaim the no-stress-tensor case. The natural stress-energy carrier in a chiral net of the C5-delta type is the modular Hamiltonian K and its boundary form rather than a local pointwise stress tensor in MZ's sense; whether the emergent IR genuinely lacks a unique local stress tensor is a property of the construction that has not been established here and is logged as open. The dimensional escape does not depend on it.

The genuine open question for Branch A is whether a $d = 2$ analogue of these results—a chiral-algebra or Virasoro-based rigidity theorem—constrains the higher-spin content of the emergent sector in a way the $d \geq 3$ theorems do not. Because $d = 2$ conformal symmetry is infinitely richer than its higher-dimensional counterpart, such an analogue would be a structurally different result, and whether one exists that bears on the emergent sector is not known. That reconnaissance has not been run. If such a result exists and bears on the emergent sector, it would close the no-stress-tensor sub-escape before any (3+1) extension is attempted, collapsing Branch A onto the dimensional escape alone; if it does not, the sub-escape stands as a second reinforcement. Either way the dimensional escape (the load-bearing defense) is unaffected. Beta remains open.

A further constraint shapes what Branch A's target could be if the programme ever reached four dimensions. The MZ and AD theorems engage only a CFT carrying an exactly conserved higher-spin current of spin greater than two; absent such a current the theorems are vacuous (the higher-spin current is their engine, not a side hypothesis). A 4D emergent CFT could therefore evade them in one of two ways. First, by carrying no exactly conserved higher-spin current at all: but a generic interacting CFT already satisfies this, so this route does not by itself select an emergence target, and the relevant pressure becomes the slightly-broken case (MZ Section 7), where higher-spin currents with order- $1/N$ anomalous dimensions still impose constraints in the large- N regime of the $O(N)$ vector models. Second, by lacking a unique local stress tensor: the disclaimed case discussed above.

The one option that is closed is a free CFT. MZ forces a CFT with exactly conserved higher-spin symmetry to free-field correlators, and a free theory is not a viable emergence target for a programme that requires interactions. Branch A’s target, if it exists, is therefore an interacting 4D theory that is either not a standard local CFT (no unique local stress tensor) or carries only approximate higher-spin structure subject to the slightly-broken constraints. Which of these the emergence programme could produce is not determined here; it is the substance of the open beta question.

Path B: Stance 2 (algebraically grounded import). If beta fails, the centralizer theorem provides a positive result: every import required by Stance 2 is now algebraically justified rather than assumed by default.

- Spacetime (flat GTG background): forced by prongs (i), (ii), and (iii).
- $SU(2)_L$ on the $\mathbb{H}$ factor: forced by the centralizer theorem ($\dim 1 < 3$).
- $SU(3)_{\text{color}}$ gauge links on a separate $\mathbb{C}^3$: forced by the double-booking obstruction.

The G-series papers provide the working realization of the spacetime import: Gauge Theory Gravity (GTG) (Lasenby et al., 1998) builds gauge-theoretic gravity in spacetime algebra on a flat Minkowski substrate, with position-gauge field $\bar{h}$ and rotation-gauge field Ω as the clean factorization of the gravitational degrees of freedom. GTG begins precisely where Stance 2 ends—it consumes the Minkowski vector space, local $\text{Spin}(1, 3)$, and smooth structure as axiomatic inputs. Stance 2’s derivation burden is exactly the substrate GTG declines to derive. The target is therefore known and precise: deliver the smooth four-dimensional space with $\text{Cl}(1, 3)$ structure at every point and $\text{Spin}(1, 3)$ acting locally, and GTG’s machinery—including the full empirical record of GR—is inherited automatically. Stance 2 is not hypothetical; it is already partially built.

Stance 2 is the algebraically grounded fallback if beta fails. It is not algebraically necessary on its own—that would require beta to be closed against—but the centralizer theorem supplies retrospective justification for the imports Stance 2 makes, rather than leaving them as unmotivated assumptions. The Furey construction (Furey, 2016) for $SU(2)_L$ on $\mathbb{H}$, in particular, is algebraically mandated by the centralizer theorem rather than merely convenient.

A concrete instance of this mandate surfaced during the Paper 3 adversarial referee review: finding B-2 noted that the candidate $T_3 = -B_{34}$ fails

to commute with $\mathfrak{su}(3)_{\text{color}}$. The centralizer theorem is the explanation— B_{34} carries $2/3$ adjoint content (Lemma 2) and cannot be color-neutral. B-2 is not a gap in Paper 3; it is the centralizer theorem asserting itself on the electroweak sector, and it forces $\text{SU}(2)_L$ off the $\text{Cl}(6)$ bivectors and onto the $\mathbb{H}$ factor exactly as Furey’s construction places it.

The two surviving paths are not symmetric. Path A (beta) is genuinely open: no mechanism is in hand for producing an S -matrix-free emergent IR from the 1+1 chiral net, and the 2D-MZ recon is an outstanding prerequisite before Branch A’s sub-escape can be evaluated on its merits. Path B (Stance 2) is already partially realized: GTG provides the spacetime import, the Furey construction provides $\text{SU}(2)_L$ on $\mathbb{H}$, and the centralizer theorem provides the algebraic justification for each choice. If beta fails, the programme does not start over—it inherits a working framework whose imports are now explained rather than assumed.

7.2 The hypercharge exception and its uniqueness

An important positive consequence of Theorem 1 is the uniqueness of hypercharge. The Standard Model contains exactly one $\text{U}(1)$ factor commuting with $\mathfrak{su}(3)_{\text{color}}$ (within the electroweak sector before symmetry breaking). The centralizer theorem shows this is not a free parameter of the model but a structural fact of the $\text{Cl}(6)$ algebra: there is room for exactly one such $\text{U}(1)$, and it is identified uniquely as $Y \propto H = \frac{3}{2} Q_{B-L}$.

7.3 The 1+1 result is unaffected

The partition no-go and the $\mathbb{H}$ dilemma constrain extensions of the existing 1+1 sector. They do not affect the 1+1 result itself, which rests on the C-series chain construction and the Tomita–Takesaki modular structure (Firestone, 2026a,b). The 1+1 sector—a massless chiral fermion with a clean Lorentzian signature—stands independently.

7.4 Connection to virtual photon compositeness

In the 1+1 sector, $d = 2$ and the transverse polarization count $d - 2 = 0$; gauge bosons are composite (Schwinger mechanism, exact) (Schwinger, 1962). This section connects two regimes: the composite-gauge-boson property of the 1+1 sector the C-series has actually built, and the activation of both obstructions in the 4D target the programme aspires to reach.

Both obstructions trace to the same source. Section 3 (the B – L reduction) and Section 4 (the partition no-go) each follow from the one-dimensional

centralizer of $\mathfrak{su}(3)$ in $\mathfrak{so}(6)$: the helicity route is closed because the unique color-neutral bivector H carries $B-L$ charge, and the spatial route is closed because the same one-dimensional centralizer cannot house $\mathfrak{so}(3)$. In 4D both walls activate together, both stemming from the centralizer dimension. The stronger statement that these two walls are literally the same wall—a single obstruction wearing two faces—is established in (Firestone, 2026e); the present paper establishes the weaker and sufficient claim that both activate and share the centralizer as their common source.

The composite gauge boson result in 1+1 is the boundary statement; the four-prong no-go is the obstruction to extending it to 4D.

8 Discussion

The central result is negative: no algebraic route to (3+1) exists within the $\text{Cl}(6)$ construction under Stance 1. This is a first-class result. A clean no-go is as valuable as a positive construction—it precisely delimits the problem that must be solved (beta) and shows what Stance 2 is forced to import and why.

The four prongs correspond to the four candidate sources of additional spatial directions in the construction, and close each in turn:

- Prong (i) closes the partition route: carving two spatial directions out of the six $\text{Cl}(6)$ internal directions.
- Prong (ii) closes the double-booking route: leaving the six directions unpartitioned and sharing generators between spatial $\mathfrak{so}(3)$ and color $\mathfrak{su}(3)$.
- Prong (iii) closes the chain-extension route: seeking additional directions from the external chain axis n (parallel Fermi-point translations, finite-dimensional per-site algebra, and discrete ladder transverse quantum number all block extension beyond 1+1).
- Prong (iv) closes the $\mathbb{H}$ -factor route: sourcing spatial rotations from the quaternion tensor factor.

These three candidate sources—the $\text{Cl}(6)$ directions, the external chain axis, and the quaternion factor—give rise to four prongs because the $\text{Cl}(6)$ reservoir admits the partition / no-partition dichotomy (prongs (i) and (ii)). The claim is completeness over these sources, not that each prong is individually irredundant in some smaller logical sense. Removing any one prong

leaves its corresponding source unaddressed, so all four are needed to close the construction as it stands. Whether a future extension of the construction introduces a fifth candidate source is a separate question, outside the present scope.

The dimension-theoretic argument at the core ($\dim \mathfrak{so}(4) = 6 < 8 = \dim \mathfrak{su}(3)$) is elementary once stated, but its consequences were not immediate because the algebraic mixing (Lemma 2) obscures any would-be clean separation. The numerical verification across all case types (nine bivector pairs in `cl6_su3_centralizer_check.py`, ten distinct spatial assignments in `cl6_spatial_direction_carving.py`, seven algebraic identities in `cl6_H_spatial_route_check.py`) confirms that the theorem is not an artifact of a particular basis choice.

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